

Technical Guide: W: Web Pipe Device Y: Clipboard Device

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Project Repo: <https://github.com/pjones1063/AspeQt-2k26>

Y: Clipboard Device (System Integration)

The Y: device provides a bi-directional bridge between the Atari 8-bit and your host computer's system clipboard (Windows, macOS, or Linux). This allows for rapid code transfer and data sharing without saving files to disk.

Usage Scenarios

Paste Code: Copy BASIC listings from a website on your PC and paste them directly into the Atari editor.

Export Data: Send program listings or variable data from the Atari to your PC's text editor.

Write to PC LIST "Y:" - Sends the current BASIC program to the PC clipboard.

Write to PC PRINT #1;"TEXT" - Sends specific text to the PC clipboard.

Read from PC ENTER "Y:" - Types the PC clipboard content into the Atari (like keystrokes).

Read from PC COPY "Y:" "E:" - Displays PC clipboard content on the screen.

Code Example: Reading & Writing

Writing to the PC Clipboard

Basic

```
10 REM Send a variable to the PC Clipboard
20 A$="Copied from Atari"
30 OPEN #1,8,0,"Y:"
40 PRINT #1;A$
50 CLOSE #1
```

Reading from the PC Clipboard Ensure you have copied text on your PC first.

Basic

```
10 DIM BUFFER$(256)
20 OPEN #1,4,0,"Y:"
30 INPUT #1,BUFFER$
40 PRINT "I found this on your clipboard:"
50 PRINT BUFFER$
60 CLOSE #1
```

W: Web Pipe Device (Thin Client Gateway)

The W: device transforms the Atari 8-bit into a "Thin Client" capable of interacting with modern web services. Instead of implementing complex network stacks (TCP/IP, SSL) on the Atari, the W: device acts as a transparent proxy. It forwards standard BASIC text commands to a backend script running on your PC via HTTP.

Multi-Channel Support (W1: - W4:)

AspeQt-2k26 supports up to 4 simultaneous connections for complex "Full Duplex" applications.

- **W1:** Primary Channel (Device \$57)
 - **W2:** Secondary Channel (Device \$56)
 - **W3:** Tertiary Channel (Device \$55)
 - **W4:** Quaternary Channel (Device \$54)
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2. On-the-Fly Mode Switching (AUX2 Flag)

By default, the W: device uses the global translation settings defined in the AspeQt Options menu. However, you can override this setting per connection using the **AUX2** parameter (the 3rd number in the OPEN command).

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Driver Logic (C++ Implementation)

The handler determines the translation mode by examining the High Byte of the AUX register. The specific logic used in the driver is as follows:

C++

```
int aux2 = (aux >> 8) & 0xFF; // Extract High Byte

if (aux2 == 1) return true;  // Force TEXT (Translate Atari $9B <-> Unix \n)

if (aux2 == 2) return false; // Force BINARY (Raw Data, No Translation)

return globalSetting;      // Default (AUX2=0)
```

Usage in BASIC

- **AUX2 = 0 (Default):** Uses the "Translate EOL" checkbox setting from the AspeQt GUI.
 - **AUX2 = 1 (Force Text):** Essential for reading text files or API responses (converts \n to ATASCII \$9B).
 - **AUX2 = 2 (Force Binary):** Essential for downloading binary data (sprites, fonts, executables) where byte values must remain unaltered.
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3. Application Examples Library

The following examples demonstrate "Hybrid Computing" by offloading processing to a Python backend.

General Prerequisites

For all examples below, you must have **Python 3.x** installed on your Host PC. **Base Install:**
pip install flask

App 1: Stock Market Ticker (New)

Fetches real-time stock quotes using the yfinance library. The Python script sanitizes the output (removing Unicode arrows/colors) to ensure safe display on the Atari.

- **Dependencies:** pip install flask yfinance
- **Python Script:** stock_ticker.py
- **BASIC File:** STOCKS.LST

App 2: Atari Newswire / RSS Reader (New)

Turns the Atari into a teletype news reader. It fetches RSS feeds (BBC, NASA), sanitizes "smart quotes," and wraps the text to 38 characters to ensure perfect formatting on the 40-column display.

- **Dependencies:** pip install flask feedparser
- **Python Script:** news_reader.py
- **BASIC File:** NEWS.LST

App 3: Gemini AI Chatbot

Turns the Atari into a client for Google's Gemini Large Language Model. You can type natural language questions ("Who is Captain Kirk?") and receive intelligent, summarized answers on the Atari screen.

- **Dependencies:** pip install google-generativeai
- **Python Script:** gemini_ai.py
- **BASIC File:** GEMINI_AI.LST

App 4: Basic Graphical Weather Station

Fetches live weather data from the Open-Meteo API for a specific location (configured in Python) and displays it on the Atari using GRAPHICS 5 (4-color mode). It draws custom pixel-art icons for Sun, Clouds, Rain, or Snow.

- **Dependencies:** pip install requests
- **Python Script:** weather.py
- **BASIC File:** WEATHER.LST

App 5: Cloud Notepad (Persistence Demo)

Demonstrates Reading and Writing to the PC's hard drive. You can type a note on the Atari, save it, turn the Atari off, and retrieve the note later. It acts as a "Cloud Drive" for text.

- **Dependencies:** flask (Standard)
- **Python Script:** cloud_note.py
- **BASIC File:** CLOUD_NOTE.LST

App 6: Text-to-Speech Synthesizer

The "Talking Atari." Any text typed on the Atari is sent to the PC, where it is read aloud using the Linux espeak engine. This demonstrates zero-latency media triggering.

- **Dependencies:** flask (Python); sudo apt install espeak (Linux System)
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- **Python Script:** tts_server.py

- **BASIC File:** TTS_SERVER.LST

App 7: Web News Scraper

Connects to a modern news website (like CBC or BBC), scrapes the HTML, strips out ads/javascript, and delivers a plain-text headline summary to the Atari.

- **Dependencies:** pip install requests beautifulsoup4
- **Python Script:** scraper.py
- **BASIC File:** SCRAPER.LST

App 8: Linux Remote Execution

Allows the Atari to execute shell commands on the Linux Host (e.g., ls -la, whoami). It includes a safety whitelist to prevent unauthorized access.

- **Dependencies:** flask (Uses standard subprocess library)
- **Python Script:** exec_linux.py
- **BASIC File:** EXEC_LINUX.LST

App 9: Windows Remote Control

Similar to the Linux version but tailored for Windows Hosts. It can launch GUI applications (Calculator, Notepad) or open URLs in the default browser.

- **Dependencies:** flask (Uses standard subprocess library)
- **Python Script:** exec_win.py
- **BASIC File:** EXEC_WIN.LST

App 10: File Upload / Telemetry

A utility to upload raw data or text files from the Atari to the PC. This can be used for saving game high scores, sensor logs, or telemetry data.

- **Dependencies:** flask
- **Python Script:** file_upload.py

- **BASIC File:** FILE_UPLOAD.LST
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Quick Dependency Install Command

To run **all** examples in this library, install the following packages on your Host PC:

Linux / macOS / Windows:

Bash

```
pip install flask requests beautifulsoup4 google-generativeai yfinance feedparser psutil
```

(Note: Linux users also need `sudo apt install espeak` for the TTS demo)